Knowledge Graph-based Retrieval-Augmented Generation System for Domain-Specific Information Extraction with Glossary-Aided Responses

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BICT Honours Degree **2025**

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Declaration

We hereby declare that this dissertation, which has been performed after registration for the degree in the Department of Information and Communication Technology Honors Degree in the Department of Information and Communication Technology at the Faculty of Technology, University of Sri Jayewardenepura, Sri Lanka and has not been previously included in a thesis or dissertation submitted to the institution or any other institution for a degree or other qualifications.

Acknowledgement

We would like to express our deepest gratitude to all those who supported us throughout the journey of this research project. This thesis is the culmination of extensive effort, collaboration, and guidance, and it would not have been possible without the contributions of several individuals and institutions. In the successful completion of our final year research thesis, we are deeply grateful to all those who provided support and guidance throughout this journey. We sincerely thank our academic supervisor, Dr. Chamara Liyanage, for his continuous encouragement, expert guidance, and valuable feedback that helped shape the direction of this research.

We would also like to express our appreciation to the Department of Information and Communication Technology, Faculty of Technology, University of Sri Jayewardenepura, for providing the necessary resources and a supportive academic environment. Our sincere thanks go to the Research Committee for approving our project and offering constructive guidance.

We are especially thankful to the Sri Lanka Tea Board for providing access to important data and resources. A special note of appreciation goes to our external supervisor, Mr. Indika Hiran Wijesinghe, Assistant Director – IT, Sri Lanka Tea Board, for his valuable collaboration and insights.

Lastly, we would like to thank our peers and colleagues for their support and helpful suggestions that contributed to the successful completion of this thesis.

Table of Contents

List of Tables

List of Figures

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
FlaskAPI	A Python-based framework for building APIs
HybridRAG	A system that combines vector-based retrieval and graph-based retrieval to enhance information extraction and generation
KG	Knowledge Graph
LangChain	A framework for managing LLM interactions and workflows
LLM	Large Language Model
MIT	A permissive software license originating at the Massachusetts Institute of Technology
Neo4j	A graph database management system
NLP	Natural Language Processing
OOAD	Object Oriented Analysis and Design
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation

React	A JavaScript library for building user interfaces

Abstract

Organizations, particularly those with extensive historical operations like government agencies or industry boards, face significant challenges managing critical information stored in extensive, unstructured historical documents such as scanned legal and regulatory files. Traditional reliance on physical archives and informal knowledge transfer leads to information retrieval inefficiencies, scalability issues, and the risk of institutional knowledge loss when experienced personnel retire. To address this, the research develops a Knowledge Graph-based Retrieval-Augmented Generation (RAG) system for domain-specific information extraction and glossary-aided responses.

The system integrates Large Language Models (LLMs) with knowledge graphs constructed by automatically extracting and structuring information from unstructured historical documents. This structured representation, stored in a graph database, facilitates semantic search and understanding. A novel element is the inclusion of a user-managed, domain-specific glossary to improve terminology consistency and interpretive accuracy for industry or legal terms. A web-based portal provides a user-friendly interface for document management and natural language querying.

The system processes user queries by retrieving relevant context from the knowledge graph and definitions from the glossary, which are then combined into a prompt for the LLM to generate accurate, contextually relevant, and domain-aware responses. Evaluation results strongly support the hypothesis: the knowledge graph-based RAG system significantly improves information retrieval efficiency and accuracy compared to manual methods. Furthermore, integrating domain-specific glossaries substantially enhances the contextual relevance and interpretative accuracy of LLM responses for specialized terms, showing a notable increase in mean answer relevancy. This system offers a scalable solution for preserving institutional knowledge and supporting informed decision-making.

1. Introduction and Background

1.1 Overview of proposed study

The proposed research aims to develop and implement an AI-powered information extraction system that leverages domain-specific knowledge through a Retrieval-Augmented Generation (RAG) framework enhanced by a knowledge graph. This system targets organisations, particularly those with extensive historical operations, such as government agencies or industry boards, where critical information is often stored in unstructured formats, including scanned legal and regulatory documents.

Many of these institutions depend heavily on traditional physical archives to retain legal, operational and procedural knowledge. However, this practice is increasingly impractical due to challenges in scalability, information retrieval inefficiencies, and the risk of institutional knowledge loss particularly when experienced personnel retire or exit the organisation, The proposed system seeks to mitigate these issues by automatically extracting, structuring, and preserving knowledge from historical documents or supporting more efficient, accurate, and informed decision-making process.

By simulating interactions with a domain expert, the system will allow users to query and retrieve information in a contextually relevant manner. The core architecture involves the integration of Large Language Models (LLMs) with knowledge graphs, enabling the generation of insightful responses that reflect both linguistic fluency and domain-specific understanding. Historical and legal documents will be processed and transformed into structured representations comprising nodes and relationships, which are stored in a graph database. This structure will facilitate semantic search and identification of patterns and relationships across documents, enhancing the interpretability of complex interconnected data.

In addition, a domain-specific glossary (like a domain-specific dictionary) will be incorporated into the system to improve the response accuracy and ensure the terminology consistency, particularly for industry-specific or legal terminology. A user-friendly, web-based document management portal will be developed to allow authorised users to interact with the system, manage uploaded documents, and update the knowledge graph as new information becomes available.

The proposed system not only improves institutional knowledge preservation but also enhances strategic planning and policy development by providing rapid access to historical precedents and data-driven insights. This integration of advanced AI techniques with structured knowledge positions the solution as a robust tool for digital transformation in knowledge-intensive domains.

1.2 Project Scope

This research project encompasses the design and implementation of comprehensive AI-powered systems that transform how organisations manage and retrieve institutional knowledge from historical, unstructured documents. Central to these initiatives is the development of an integrated solution that combines advanced information retrieval techniques, semantic understanding, and domain-specific knowledge representation to support informed decision making.

1.2.1 Development of a Web-Based Document Management Portal

A modern, secure, and scalable web-based portal was developed to serve as the central interface for managing the organisation's historical document repository. Authorised users have been provided with functionalities to upload, view, and chat with the documents. This portal has replaced manual, paper-based archival methods, improving efficiency, enhancing document traceability, and ensuring long-term digital preservation of critical knowledge.

1.2.2 Construction of a Domain-Enriched Knowledge Graph

A structured knowledge graph was constructed from a corpus of historical and legal documents. Textual content was segmented into meaningful information units (nodes), and semantic relationships between these nodes were identified and encoded. A domain-specific glossary was incorporated to enhance the semantic accuracy of the graph, enabling context-aware queries and precise information extraction.

1.2.3 Integration of Advanced AI Technologies

The system integrates Large Language Models (LLMs), Langchain, and a Graph database (Neo4j) to form a knowledge graph-based Retrieval-Augmented Generation (RAG) architecture. User queries are interpreted through natural language processing techniques, with key entities and relationships extracted and aligned against the knowledge graph. The system then synthesises responses by combining structured graph-based retrieval with the generative capabilities of LLMs.

1.2.4 Design of Intuitive User Interfaces.

A user-centric, responsive interface has been designed and implemented to facilitate seamless interaction with the system. Users can enter natural language questions, explore document content, and receive accurate, well-contextualised answers. The interface supports users of varying technical backgrounds and accessible user experience, thereby increasing adoption and usability across organisational roles.

1.2.5 Integration of Domain-Specific Glossary

A specialised glossary containing domain-specific terminology was developed and integrated into the knowledge graph and retrieval components. This glossary has improved the system's ability to interpret domain-specific terminologies, enhancing both query understanding and response accuracy. It ensures that domain-specific terms are consistently recognised and sent to the LLM with context for answer generation

1.2.6 Empowerment of Users through Interactive Knowledge Retrieval System

The system empowers users through an Interactive Knowledge Retrieval System that enables natural language interaction with institutional knowledge. By connecting unstructured historical records with structured insights, it addresses challenges like knowledge loss due to personnel turnover and fragmented documentation. This facilitates informed decision-making by providing users with timely and contextually relevant information.

1.2.7 Scalability and Maintenance Considerations

The system architecture was designed with scalability and maintainability as core principles. It supports the incremental addition of documents, knowledge graph expansion, and increasing user query loads without compromising performance. Additionally, the system allows for modular updates, bug fixes, and integration of new functionalities, ensuring long-term adaptability to evolving organisational requirements and technological advancements.

1.2.8 Collaboration and Ethical Compliance

Throughout the research, strict adherence to ethical standards governing the handling of sensitive and proprietary data was maintained. Close collaboration with the participating organisations ensured legal, ethical, and institutional protocols were followed. Necessary permissions and data access clearance were obtained prior to system deployment, ensuring compliance with data protection regulations and ethical research practices.

1.3 Motivation to Study

The primary motivation behind this study stemmed from the urgent need to modernise the Sri Lankan Tea Board's method of managing and retrieving historical legal and regulatory documents. Since 1976, the Sri Lanka Tea Board has accumulated a vast collection of documents encompassing critical decisions, industry guidelines, and regulatory acts that form the backbone of operational and policy-related decision-making within the tea sector. Traditionally, access to this body of knowledge was dependent on manual file systems and the institutional memory of experienced personnel.

This reliance on physical archives and informal knowledge transfer mechanisms presented several significant challenges. The gradual retirement or turnover of experienced staff members posed a serious risk of institutional knowledge erosion, leading to decreased continuity and potential inefficiencies in governance and regulatory compliance. Furthermore, the increasing complexity of the tea industry and its evolving legal frameworks made it progressively difficult for new and existing staff to efficiently access relevant historical information.

These limitations highlighted a clear need for an intelligent, sustainable solution that could preserve institutional memory while enhancing operational efficiency. The study was thus motivated by the goal of designing a robust AI-powered system capable of extracting, organising and retrieving contextually relevant insights from decades of documentation. By simulating the expertise of domain specialists and integrating structured knowledge representations, the system aimed to bridge the existing knowledge gap, support informed decision-making, and ensure long-term accessibility of critical information for future stakeholders within the industry.

1.4 Rationale and Justification

The rationale for this study was rooted in the critical need for organisations with rich institutional histories, such as the Sri Lankan Tea Board, to modernise the management of their historical document repositories. Across many sectors, especially those governed by regulatory decisions, policy implementations, and industry-specific directives. However, as the volume of documentation expands and experienced personnel retire or exit the organisation, retrieving relevant information becomes increasingly challenging. This contributes to a gradual erosion of institutional memory, which in turn hampers effective decision making, introduces operational delays, and risks the loss of crucial historical insights.

This study was justified by its potential to address these limitations through the implementation of a novel AI-powered system. Specifically, the research proposed a Knowledge Graph Based Retrieval-Augmented Generation (RAG) framework that integrates knowledge graph retrieval methods. This graph-based approach was designed to ensure precise, context-aware access to historical information by simulating expert-level understanding of the domain. By transforming unstructured documents into an intelligent linked knowledge graph, the system enabled users to retrieve and interpret historical content with greater accuracy and efficiency.

The justification further extended to the long-term value this system provided in preserving and operationalising institutional knowledge. The developed solution not only facilitated streamlined access to historical data but also ensured that insights drawn from past documents could inform present and future decisions. Through enhanced information retrieval, improved decision support, and promotion of continuity across organisational transitions, the study offered a scalable and adaptable technological intervention. Ultimately, the outcomes of this research contributed to closing the gap between traditional document storage methods and modern AI-driven knowledge management systems, thereby reinforcing the strategic capacity of institutions like Sri Lankan Tea Board.

1.5 Research problem

Organisations with long-standing operational histories often encounter significant challenges in managing and retrieving critical information embedded within large volumes of historical documents. These documents frequently encapsulate key regulatory decisions, institutional knowledge, and context-specific insights that are indispensable for informed and strategic decision-making. Despite their importance, many such organisations, particularly in the public and regulatory sectors, continue to rely on outdated, manual systems for storing and accessing these archives. These traditional practices are not only time-consuming and inefficient but also highly susceptible to information fragmentation and loss.

The situation becomes more critical as experienced personnel, who possess tacit domain knowledge, retire or leave the organisation. Without a systematic knowledge preservation and transfer mechanism in place, institutional memory weakens over time, leaving newer staff without the contextual understanding required to interpret past decisions or historical data. This knowledge gap compromises the quality and speed of decision-making and diminishes the organisation's ability to respond effectively to emerging challenges.

These issues underscore the urgent need for a modern, intelligent system capable of automatically managing, retrieving, and contextualising historical data. Such a solution would enable organizations to preserve vital domain expertise, improve access to context-rich information, and ensure continuity of decision-making. The problem is especially acute in domains like the tea industry, where regulatory decisions and historical documentation date back several decades and remain central to ongoing operations. Addressing this problem requires not just digitisation, but a sophisticated AI-powered approach that can understand, structure, and retrieve information with expert-level accuracy.

1.6 Research Objectives, Hypothesis and Question

1.6.1 Main Objective

The primary objective of this research is to design and develop a Knowledge Graph-based Retrieval Augmented Generation (RAG) system that enhances an organisation's ability to manage, retrieve, and utilise historical documents effectively. By leveraging state-of-the-art technologies such as Large Language Models (LLMs), knowledge graphs, and domain-specific glossaries, the system aims to support informed decision-making through the provision of accurate, contextually relevant information derived from decades of institutional records and regulatory documentation.

1.6.2 Specific Objectives

To achieve the overarching goal, the study has outlined the following specific objectives.

- To design and implement a web-based document management portal that facilitates efficient uploading, viewing, validation, and deletion of documents by authorized personnel, ensuring a centralized and structured repository for historical records.
- To extract structured knowledge from unstructured historical documents by converting document text into knowledge graph representation and enabling enhanced organization, semantic search, and retrieval.
- To develop an intuitive and user-friendly interface that allows users to interact with the system in natural language, thereby reducing the learning curve and ensuring usability across varying levels of technical expertise.

- To incorporate a domain-specific dictionary or glossary into the system to ensure correct interpretation of industry-specific terminology, thereby enhancing the semantic accuracy and relevance of LLM-generated responses.

1.6.3 Hypotheses

The research is guided by the following hypotheses

- H1: The implementation of a knowledge graph-based RAG system significantly improves the efficiency of information retrieval compared to traditional manual methods of document management.
- H2: The integration of a domain-specific glossary enhances the contextual relevance and interpretative accuracy of responses generated by the LLM. This leads to more relevant answers by significantly improving the LLMs's contextual understanding.

1.6.4 Research Questions

This research seeks to answer the following questions

1. How can a Knowledge Graph based RAG system be effectively utilized to manage and retrieve information from historical documents within an organization?
2. What is the measurable impact of implementing the Knowledge Graph-based RAG system on the efficiency and accuracy of information retrieval compared to existing manual practices?

3. In what ways does the integration of domain-specific glossary influences the performance of the LLM in generating accurate and contextually relevant responses?
4. What are the key challenges in implementing and deploying such a system in a real-world organizational environment, and how can these be addressed to ensure its successful adoption, scalability, and long-term sustainability?

1.7 The conceptual, theoretical and practical significance of the research

This research holds multifaceted significance, conceptual, theoretical, and practical within the realms of knowledge management, artificial intelligence, and organizational decision-making.

Conceptually, the study explores the innovative application of Knowledge Graph Based Retrieval Augmented Generation (RAG) models in the domain of institutional knowledge management. It contributes to a growing body of knowledge that examines how advanced AI systems can be customized to address the unique challenges faced by organizations with extensive historical archives. By focusing on a real-world use case, the Sri Lanka Tea Board’s this research demonstrates how a knowledge graph based RAG architecture can bridge the gap between unstructured textual data and structured organizational knowledge, thereby extending conceptual frameworks in AI-assisted archival retrieval and domain-specific information systems.

Theoretically, the research advances the discipline of intelligent information retrieval by proposing and validating a novel approach that integrates knowledge graphs and Large Language Models (LLMs). This integrated architecture offers a more holistic understanding of how semantic relationships and contextual embeddings can be jointly leveraged to extract meaning from complex, historically dense document corpora. The theoretical contribution lies in its potential to redefine the way contextual relevance and accuracy are achieved in long-term document repositories, particularly in regulated and domain-specific environments.

Practically, this research addresses an urgent need for organizations to preserve and utilize institutional knowledge in an efficient and scalable manner. Traditional methods of recordkeeping and manual search are often inefficient in the face of growing data volumes and gradual loss of experienced personnel. The system developed through this study offers a scalable, AI-driven solution that supports day-to-day decision-making, enhances institutional memory, and ensures that valuable regulatory and operational knowledge remains accessible. Furthermore, it demonstrates a replicable model that can be extended to other sectors such as law, education, healthcare, and public administration, where historical context and data continuity are equally critical.

In essence, this research not only contributes new knowledge to academic fields but also delivers a tangible technological solution to a pervasive organizational challenge, thereby reinforcing its relevance across both scholarly and real-time domains.

1.8 Expected Outcomes / Anticipated results

This research is anticipated to deliver a fully functional and Knowledge Graph based Retrieval Augmented Generation (RAG) system, that will transform the way organizations manage, access, and utilize historical legal and regulatory documents. The system will serve as a strategic knowledge infrastructure, offering long-term benefits to institutional decision-making, operational continuity, and digital transformation.

1.8.1 Enhanced Information Retrieval

The proposed system is expected to significantly improve efficiency, speed, and accuracy of information retrieval by combining semantic search with graph-based contextualization. Users will be able to obtain precise and contextually relevant insights from decades of archival documents with minimal effort, thereby eliminating the delays and inconsistencies commonly associated with manual searches or reliance on domain experts.

1.8.2 Preservation and Utilization of Institutional Knowledge

By structuring domain-specific insights extracted from historical documents into an integrated knowledge graph, the system will preserve critical institutional memory. This will mitigate the risk of knowledge loss due to staff turnover or retirement and ensure that organizational knowledge remains accessible and actionable for future employees and decision-makers.

1.8.3 Improved Decision-making Capacity

The intelligent querying capabilities of the Knowledge Graph Based RAG system—driven by a Large Language Model (LLM)—will provide stakeholders with data-driven, historically-informed recommendations. This will be particularly valuable in addressing recurring regulatory, legal, and operational scenarios that depend on precedents and past decisions. As a result, decision-making will become more consistent, informed, and aligned with institutional history.

1.8.4 User-friendly and Inclusive Interface

A key expected outcome is the development of an intuitive, user-friendly web interface that allows users across different roles, regardless of their technical background, to seamlessly interact with the system. This will promote widespread adoption within the organization, reduce the learning curve, and ensure that the benefits of the system are accessible to both technical and non-technical users.

1.8.5 Academic and Practical Contributions

The research will provide valuable contributions to both academic literature and practical implementations in the fields of intelligent document management, knowledge representation, and AI-assisted information retrieval. Academically, it will offer a case study on the integration of RAG architectures with domain-specific knowledge graphs. Practically, it will serve as a scalable, replicable solution for other government agencies and industry bodies facing similar document management and knowledge preservation challenges.

1.8.6 Long-Term Sustainability and Scalability

Designed with maintainability and future scalability in mind, the system is expected to provide a long-term, adaptable solution. It will support continuous integration of new documents, evolving domain vocabulary, and technological upgrades. This ensures that organizations can grow with the system, maintaining relevance and strategic advantage in a constantly changing regulatory and operational environment.

2. Literature Review

Organizations across diverse industries routinely grapple with the challenges of extracting accurate, context-rich information from extensive repositories of legal, technical, and operational documents, many of which exist in unstructured formats such as scanned PDFs. Traditional retrieval methods, relying on keyword matching or basic statistical models, frequently falter when confronted with domain-specific terminology, complex sentence structures, and the latent semantic relationships that span across large document sets (Zhao, Pan, & Yang, 2020). To overcome these limitations, recent research has coalesced around Knowledge Graph based Retrieval-Augmented Generation (RAG) architectures that integrate the structured relational power of knowledge graphs with semantic flexibility (Sarmah et al., 2024).

Zhao et al. (2020) were among the first to demonstrate how knowledge graphs could elucidate hidden relationships within domain specific texts. By applying a TextCNN-based topic extraction model to technical documents and storing the results entities and relations in Neo4j, they showcased significant gains in retrieval accuracy compared to baseline keyword approaches. Their work established that graph databases can surface implicit semantic links, such as functional dependencies or hierarchical component relationships, that simple statistical models overlook Zhao, Pan, & Yang, 2020). However, Zhao's implementation was confined to well-structured technical manuals, leaving open the question of how knowledge graphs might perform on highly heterogeneous, unstructured corpora like scanned legal contracts or operational logs.

Addressing this gap, Sarmah et al. (2024) introduced a Hybrid RAG framework combining two complementary modules: VectorRAG, which encodes unstructured text segments into high-dimensional embeddings, and GraphRAG, which navigates structured metadata stored in a knowledge graph. In benchmark tests on mixed-format datasets, their hybrid model outperformed monolithic RAG systems by 8-12% in both precision and recall, especially when querying the required synthesizing information across metadata fields and free text passages (Sarmah et al., 2024). This early validation of the hybrid approach underscored its potential for real-world applications, yet Sarmah at al. did not explore mechanisms for automating model configuration or for integrating domain-specific lexicons to disambiguate specialized terms.

To minimize manual tuning and enhance adaptability, Kim et al. (2024) proposed AutoRAG, an automated model-selection algorithm. AutoRAG conducts a greedy search over possible RAG sub-model combinations, VectorRAG alone, GraphRAG alone, or their hybrid, and evaluates each configuration against small probing queries. By examining metrics such as term-distribution skewness and average graph-connectivity scores, AutoRAG selects the optimal setup for a new document corpus without human intervention. In their experiments, this method achieved comparable retrieval performance to an expert-tuned system while slashing deployment time over 40% (Kim et al., 2024).

Complementing AutoRAG, Yu, He, and Glass (2024) developed AutoKG, a framework for automated knowledge graph construction that leverages Open Information Extraction (OpenIE) to identify candidate entity, relation triples and BERT-based embeddings to cluster and refine those triples into a coherent graph. AutoKG's end-to-end pipeline reduced manual annotation overhead by more than 60% and proved robust across disparate domains, including technical support ;logs and compliance documents, where labeled training data was scarce (Yu, He, & Glass, 2024). By combining AutoRAG's model selection capabilities with AutoKG's rapid graph construction, organizations can potentially deploy fully automated RAG systems within days rather than months.

Beyond automation, agent orchestration has emerged as a strategy for scaling RAG systems in enterprise environments. Jeong (2024) presented a LangChain-based multi-agent RAG architecture in which discrete "agents" specialize in tasks such as document retrieval, graph traversal, query rewriting, and final answer generation. A central orchestrator routes each subquery to the most appropriate agent, drawing on both Google and OpenAI APIs, to optimize for relevance, latency, and cost. In enterprise pilot studies, Jeong's system reduced end-to-end query latency by 25% and increased ser satisfaction in question-answering tasks by 18% compared to monolithic RAG deployments (Jeong, 2024).

While automation and orchestration streamline deployment, effective retrieval in specialized domains also demands a context-aware dictionary to handle rare or polysemous terms. Hu et al. (2024) addressed this by leveraging large language models (LLMs) to bootstrap knowledge graphs from unlabeled text. Their pipelines use an LLM to annotate entity types and relation classes in raw sentences, which are then assembled into a provisional graph that undergoes human-in-the-loos refinement. Although Hu et al. did not explicitly incorporate named dictionaries, their approach highlights how LLMs can compensate for scarce annotation by generating high-quality candidate structures (Hu et al., 2024).

In parallel, Omrani, Hosseini, and Hooshangirani (2024) focused on granular context modeling within Hybrid RAG frameworks. They introduced Sentence-Window RAG, which divides long documents into overlapping text windows to maintain local coherence, and Parent-Child RAG, which links section-level summaries (parent nodes) to individual sentences (child nodes). On a benchmark of mixed-format legal and compliance documents, their dual-strategy model delivered a 12% improvement in contextual relevance scores over baseline hybrids (Omrani et al., 2024).

Doan et al. (2024) demonstrated the domain-agnostic potential of hybrid retrieval in healthcare, especially integrating knowledge graphs. By embedding structured patient metadata and unstructured clinical notes into a unified vector space, while simultaneously consulting a knowledge graph of medical ontologies, they achieved a 30% increase in correct context retrievals for complex diagnostic queries, compared with pre-vector-only baselines. Their findings suggest that hybrid RAG approaches can generalize across knowledge-intensive fields (Doan et al., 2024).

Despite these advances, several critical gaps remain unaddressed. First, while AutoKG and Hu et al.'s LLM-bootstrapping pipeline alleviate annotation bottlenecks, neither explicitly integrates curated domain-specific dictionaries that could further disambiguate rare terms during both graph construction and vector embedding. Second, AutoRAG's greedy search optimizes for speed and general retrieval metrics but may overlook low-frequency yet high-importance vocabulary, particularly when deployment corpora contains specialized jargon. Third, comprehensive evaluations on truly heterogeneous corpora, combining scanned PDFs, structured metadata, embedded tables, and domain glossaries, are still scarce (Banerjee et al., 2024; Fang, Meng, & Macdonald, 2024; Edwards, 2024; Su et al., 2024).

Our proposed research directly addresses these gaps through three innovations:

1. Automated Knowledge Graph + Model Pipeline: We will integrate Knowledge Graph and AutoRAG methodologies to build and select optimal RAG configurations with minimal human oversight (Kim et al., 2024; Yu, He, & Glass, 2024).
2. Dictionary-Enhanced Representations: A context-aware dictionary module will enrich the knowledge graph by labeling nodes with precise definitions, aiding both retrieval and disambiguation.

Performance will be evaluated across multiple dimensions, including retrieval precision, response coherence, dictionary-driven disambiguation accuracy, and end-to-end system latency. By fusing automated RAG optimization with dictionary-enhanced knowledge graph representations, this research aspires to establish a next-generation RAG framework purpose-built for complex, heterogeneous document ecosystems. The resulting system will not only accelerate information extraction but also dramatically enhance interpretability and domain relevance—enabling organizations to unlock actionable insights from vast, unstructured archives with a level of precision, scalability, and contextual intelligence previously unattainable.

3. Methodology

3.1 Proposed Experiments/Investigations and Techniques

To achieve the main and specific objectives of this research, a series of carefully designed experimental procedures and technical methods have been proposed. These are aimed at building an intelligent Knowledge Graph-based Retrieval-Augmented Generation (RAG) system that not only supports efficient retrieval of domain-specific information from historical documents but also delivers glossary-aided responses to enhance contextual understanding.

The key investigations and techniques include:

3.1.1 Knowledge Graph Creation from Historical Documents

A pipeline was designed to automatically extract structured knowledge from decades-old unstructured documents. This involves document ingestion, text extraction, chunking, and entity-relationship extraction using LLM-based tools such as LangChain's LLMGraphTransformer. The extracted entities and relationships are then used to construct a semantically rich knowledge graph in a graph database.

3.1.2 Glossary-Aided Contextual Enhancement

A novel aspect of this research is the integration of a user-managed, domain-specific glossary to enhance the interpretive accuracy of responses. This glossary is dynamically queried using fuzzy matching techniques to identify and retrieve definitions relevant to user queries. This method ensures the LLM understands domain-specific terminology, a challenge often overlooked in traditional RAG systems.

3.1.3 Context Retrieval

User questions are processed using an LLM to extract key entities, which are then used by a Graph Retriever to query the knowledge graph for related nodes and relationships. Simultaneously, glossary terms related to the query are identified and retrieved. Both the graph-based context and glossary definitions are combined into a structured prompt.

3.1.4 Response Generation via Prompt Engineering

The structured prompt—consisting of the user's question, retrieved graph context, and glossary definitions—is passed to an LLM to generate a contextually accurate and domain-aware response. This experiment evaluates how domain-aware responses differ in quality and relevance when glossary definitions are included versus when they are not.

3.1.5 Comparative Evaluation of System Effectiveness

To validate the system's effectiveness, experiments will be conducted comparing:

- Retrieval efficiency (response time and accuracy) with and without the knowledge graph.
- Quality and contextual relevance of generated responses with and without glossary integration.Metrics such as precision, relevance scores, and user satisfaction (from evaluative surveys) will be used to measure improvements.

3.2 Justification of Methodology to Realize Objectives

3.2.1 Realising the Main Objective

The main objective is to design and develop a Knowledge Graph-based Retrieval-Augmented Generation (RAG) system that improves access to and understanding of information embedded in historical documents. This is achieved through the integration of three core technologies:

1. Large Language Models (LLMs): Used for both triple extraction and response generation, LLMs enable the system to interpret complex user queries and generate coherent, context-aware answers.
2. Knowledge Graphs: Created using extracted entities and relationships, knowledge graphs offer a structured and semantically rich representation of historical content, allowing efficient retrieval and contextual linking of related information.
3. Glossary Integration: A novel component of the system, the domain-specific glossary enhances the LLM's understanding of industry-specific terms, improving the relevance and accuracy of generated responses.

3.2.2 Realising the Specific Objectives

- Document Management Portal Development this is addressed by building a web-based system that enables authorized users to upload, view, and manage historical documents. The portal serves as the entry point to the RAG pipeline, ensuring that all documents are centralized and available for processing.
- Structured Knowledge Extraction from Documents this objective is fulfilled through the use of the LangChain "LLMGraphTransformer", which enables

automated extraction of entities and relationships from unstructured text. The resulting graph structure in Neo4j supports enhanced semantic search and contextual retrieval.

- Natural Language Interface for User Interaction by integrating LLMs into the query processing layer, users can interact with the system using natural language without requiring technical knowledge. This makes the system accessible and usable across a broad range of users within the organization.
- Incorporation of Domain-Specific Glossary the glossary is managed through the web application, allowing users to update or add new domain terms. Fuzzy matching is applied during query processing to retrieve relevant glossary definitions, which are then included in the prompt sent to the LLM. This improves the interpretive accuracy of the system, especially in domains with specialized terminology.

3.3 Ethical Considerations and Organisational Consents

3.3.1 Ethical Clearance

Although the original intention was to use confidential historical documents from the Sri Lanka Tea Board, access restrictions have prevented their use in the experimental and testing phases of this research. To ensure ethical compliance and avoid the use of sensitive or proprietary information, an alternative dataset comprising publicly available clinical trial reports has been used. This dataset is licensed under the MIT License, allowing free use for academic and research purposes.

All experimentation in this research is therefore based on publicly available data, and no personally identifiable or sensitive information is processed. The study adheres to the ethical guidelines outlined by the university, particularly those related to data privacy, responsible AI usage, and transparent methodology.

3.3.2 Collaboration with the Sri Lanka Tea Board

While the Tea Board’s internal documents were not used in the experimental phase, the project continues to receive advisory support from the Assistant Director of Information Technology at the Sri Lanka Tea Board, who serves as the external supervisor. This collaboration ensures that the system’s design and functionality remain aligned with real-world institutional needs, even though the testing data originates from an alternative source.

3.3.3 Data Security and Privacy

To simulate real-world usage scenarios where document confidentiality is critical, the developed system includes several data security mechanisms. These include:

- Access control within the document management portal to restrict document upload and viewing to authorized users only.
- Encryption of stored documents and communication between the web application and backend services.
- Audit logging to monitor document access and activity within the system.

3.4 Functional and Non-Functional Requirements

3.4.1 Functional Requirements

Document Upload and Management

- Authorized users should be able to upload domain-specific documents through a secure document management portal.
- Users must have options to view, validate, and delete uploaded documents as necessary.

Text Extraction and Preprocessing

- The system must extract text from uploaded documents (PDF format) and split the content into smaller text chunks for efficient processing.

Entity and Relationship Extraction

- A Large Language Model (LLM) should process each chunk to extract named entities and relationships using the LLMGraphTransformer from the LangChain framework.

Knowledge Graph Construction

- Extracted entities and relationships must be stored in a Neo4j graph database, with nodes representing entities and edges representing relationships.

User Query Interface

- Users must be able to submit questions in natural language through a web-based interface.

Graph-Based Context Retrieval

- The system should extract key entities from the user query and retrieve related nodes and relationships from the knowledge graph using a Graph Retriever component.

Glossary Management and Integration

- The system must support glossary management where authorized users can add, update, or delete domain-specific terms and definitions.
- During query processing, glossary definitions should be fetched and integrated into the context using fuzzy matching against user input.

Prompt Construction and Response Generation

- The system must construct a structured prompt combining the user's query, retrieved context, and glossary definitions, which will be passed to the LLM for final response generation.

Answer Presentation

- The generated response should be presented clearly to the user through the same web interface.

3.4.2 Non-Functional Requirements

Performance

- The system should return an LLM-generated response within an acceptable time frame (ideally under 5 seconds for common queries).

Scalability

- The system should be able to scale horizontally to handle an increasing volume of documents and user queries.

Security

- Document access must be restricted to authenticated users.
- Stored documents and glossary entries must be protected through access controls and secure storage mechanisms.

Usability

- The web application should have an intuitive and user-friendly interface requiring minimal training.
- Prompts, feedback messages, and form validation should support accessibility and smooth navigation.

Maintainability

- Code and configurations should follow modular design principles to support long-term maintainability and enhancements.

Reliability

- The system should ensure consistent uptime and gracefully handle exceptions during document processing, retrieval, or LLM failures.

3.5 Specific Use of Technologies and Techniques

The proposed system is a Knowledge Graph-based Retrieval-Augmented Generation (RAG) System designed specifically for Domain-Specific Information Extraction with Glossary-Aided Responses. It combines modern natural language processing (NLP) techniques with structured knowledge representation to enhance retrieval accuracy and contextual relevance. The integration of a domain-specific glossary ensures that the system's responses remain aligned with key terminologies and definitions. To realize this, a set of well-chosen technologies and techniques are employed, each playing a specific role in the pipeline. The applied nature of this research demands a hands-on, implementation-driven approach supported by the following components.

3.5.1 Large Language Models (LLMs)

- Use Case: Triple extraction, entity extraction in queries, and final answer generation.
- Technique: Prompt engineering is used to guide LLMs (Llama 3.1 & Gemma 3) in extracting relevant information from document chunks and constructing knowledge triples.
- Rationale: LLMs demonstrate strong zero-shot and few-shot performance in understanding domain-specific language and complex sentence structures, making them ideal for unstructured text understanding.

3.5.2 LangChain and LLMGraphTransformer

- Use Case: Automating the transformation of unstructured textual data into a structured knowledge graph representation.

- Technique: LangChain provides a flexible framework to orchestrate interactions between Large Language Models (LLMs) and other components. Within this framework, the LLMGraphTransformer module leverages LLM capabilities to extract entities and relationships from document chunks, converting them into graph-compatible formats. LangChain also handles prompt management, chaining of processing steps, and integration with external data stores.
- Rationale: This approach eliminates the heavy reliance on manually crafted rules and traditional NLP pipelines, enabling the system to adapt quickly to different domains and document types by leveraging the contextual understanding of LLMs. The modular design of LangChain supports extensibility and simplifies the integration of new processing components.

3.5.3 Neo4j (Graph Database)

- Use Case: Storing and querying the structured knowledge extracted from documents.
- Technique: Nodes represent domain-specific entities, while edges capture the relationships between them. Cypher queries are used to retrieve relevant subgraphs based on user queries.
- Rationale: Neo4j offers a robust, scalable graph data model ideal for representing semantic relationships and supporting complex query patterns.

3.5.4 Glossary Integration with Fuzzy Matching

- Use Case: Providing glossary-aided contextual support during response generation.

- Technique: Fuzzy string matching (e.g., using Levenshtein Distance or RapidFuzz) is applied to match user queries or extracted terms with glossary entries stored in a relational database.
- Rationale: Glossary integration ensures that the generated answers remain aligned with domain-specific definitions and terminologies, thereby improving accuracy and trustworthiness.

3.5.5 React.js (Frontend)

- Use Case: User interface for uploading documents, managing glossary terms, and submitting queries.
- Technique: Built with React components to offer a responsive, interactive, and user-friendly experience.
- Rationale: Modern front-end architecture enables seamless interaction with the backend while supporting dynamic data rendering.

3.5.6 Python and Flask (Backend APIs)

- Use Case: Orchestrating the data flow between the frontend, knowledge graph (Neo4j), glossary database (MongoDB), LLM APIs (Groq/DeepInfra), and Azure Blob Storage.
- Technique: RESTful endpoints built with Flask handle core operations such as document upload, metadata storage, triple extraction, knowledge graph updates, glossary term management, and query processing. These endpoints act as the bridge between user actions and backend logic, ensuring smooth, asynchronous interactions.

- Rationale: Flask is a lightweight and flexible Python web framework that supports rapid API development and easy integration with external services. It is well-suited for applications requiring seamless connectivity between data processing components and AI services, offering simplicity, modularity, and scalability for backend development.

3.5.7 Secure File Handling and Preprocessing (PDFs)

- Use Case: Extracting and preprocessing text from PDF documents.
- Technique: Python libraries like PyMuPDF is used to extract clean, structured text from scanned or digitally generated PDFs.
- Rationale: Proper preprocessing is critical to ensure LLM input quality and reduce noise in triple extraction.

4. Timeline

The successful completion of this research project relies on a well-structured timeline that ensures all key tasks are systematically addressed. The timeline was designed to guide the project from initial stages of topic selection and literature review through to the final stage of thesis submission and presentation. The structured approach enabled continuous progress monitoring and timely adjustments to keep the project aligned with objectives. As of now, all planned phases up to the thesis writing have been completed.

The following timeline diagram outlines the key milestones and tasks undertaken over the research project, providing a clear roadmap that supports the achievement of the research objectives.

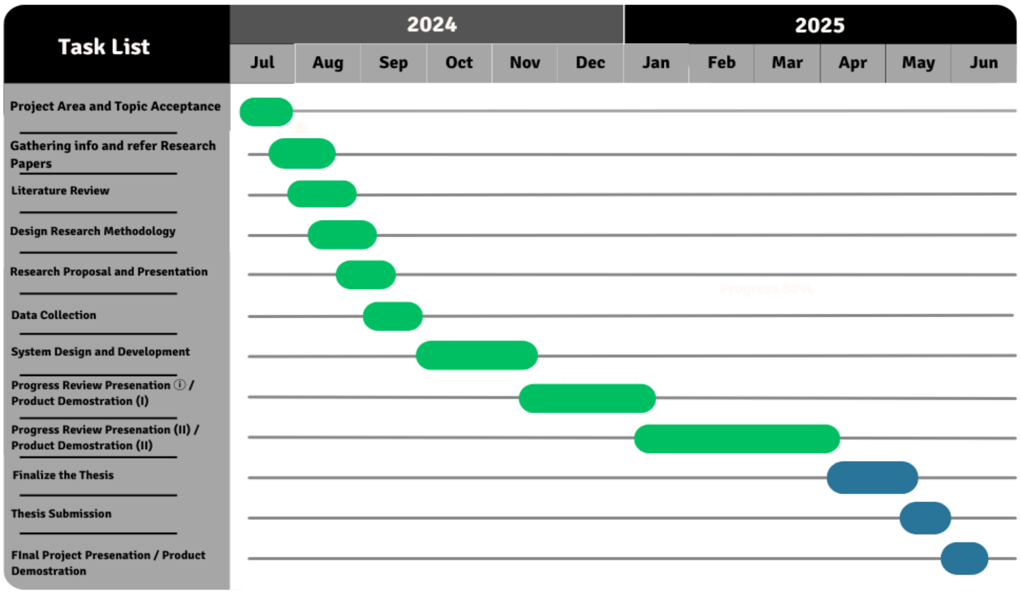

Figure 1: Timeline

5. Experimental Plan/ Project Design/ System Design Overview

5.1 System Design Overview

The system architecture for this project is designed around a modular, multi-tiered architecture that seamlessly integrates a web-based user interface, a robust backend service layer, and graph-based knowledge representation model. It facilitates intelligent information extraction and retrieval using a Retrieval-Augmented Generation (RAG) pipeline, optimized for domain-specific document understanding. The frontend, developed using React, provides users with a Document Management Portal that supports secure upload, search, viewing, and deletion of documents. Uploaded documents are securely stored in Azure Blob Storage, ensuring both data integrity and confidentiality.

In parallel, the system utilizes MongoDB for storing documents metadata. Each uploaded document is associated with a unique hash value, computed per document; also it prevents duplicate uploads. This mechanism ensures storage efficiency and data accuracy across the platform. The system also maintains a chat history collection within MongoDB, where all chat histories are persistently stored. This stored chat history is later referenced during answer generations to improve the continuity and contextual accuracy of responses.

On the backend, a Flask-based API, which enhanced with essential python libraries such as LangChain, LangChain-Groq, Neo4j, PyMuPDF, PaddleOCR, DeepInfra, Groq, RapidFuzz, Flask-CORS, PyMongo and other python libraries. These libraries handle document processing, natural language understanding, and advanced LLM integration. Upon document upload, the system extracts text, identifies entities and relationships, and updates the domain-specific Knowledge Graphs, which is maintained in Neo4j. This graph-based structure enables a semantically rich representation of information through interconnected nodes and relationships.

In addition to the knowledge graph, the system includes a Glossary Management Module, allowing users to maintain a curated dictionary of industry-specific terms. When a user submits a question through the web interface, the system performs entity extraction on the query using LLM via LangChain, followed by semantic search on the Knowledge Graph to retrieve relevant contexts. If the question contains domain-specific terminology that may not be directly understood by the LLM, the system performs fuzzy matching against the glossary database to retrieve precise definitions. These glossary definitions, combined with the original query, relevant graph context, and related chat history, are compiled into structured prompt and sent to the LLM for context-aware response generation.

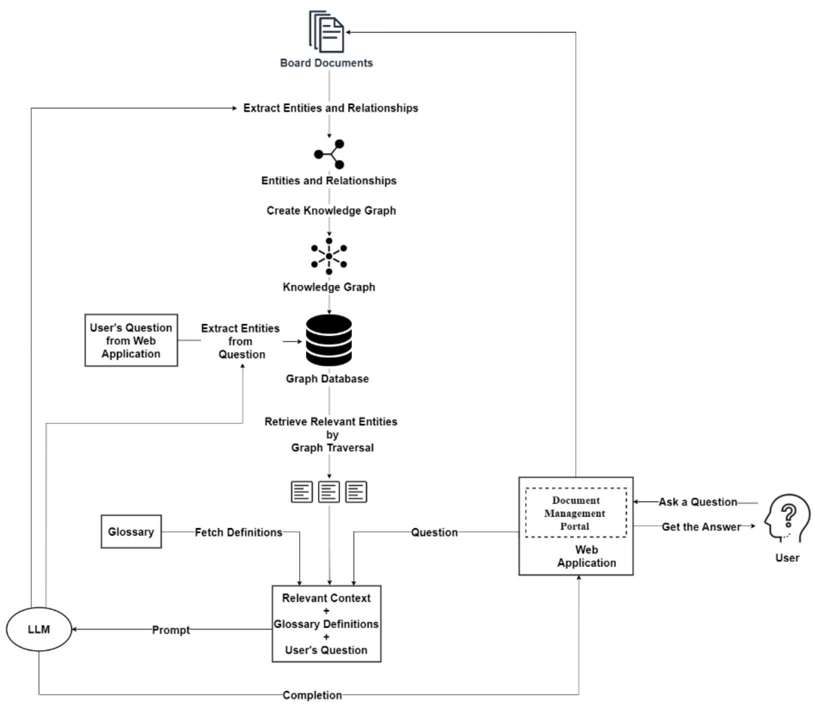

Figure 2: System design

This architecture ensures a scalable, secure, and intelligent platform by combining state-of-the-art AI techniques, cloud-native storage, robust data management with MongoDB, and flexible knowledge graphs. The system supports real-time question answering, efficient document and glossary management, and domain-specific language modelling, making it highly effective for advanced information retrieval in specialized domains.

5.2 Frontend Architecture

The frontend of the system is designed using React.js, a component-based JavaScript library that enables the creation of dynamic and responsive user interfaces. It serves as a primary interface through which users interact with the system. The frontend comprises multiple interface modules that support key functionalities of the system, including secure document management, glossary maintenance, question submission, and viewing of AI-generated responses. Additionally, it includes multiple custom themes to enhance visual experience and allow user personalization.

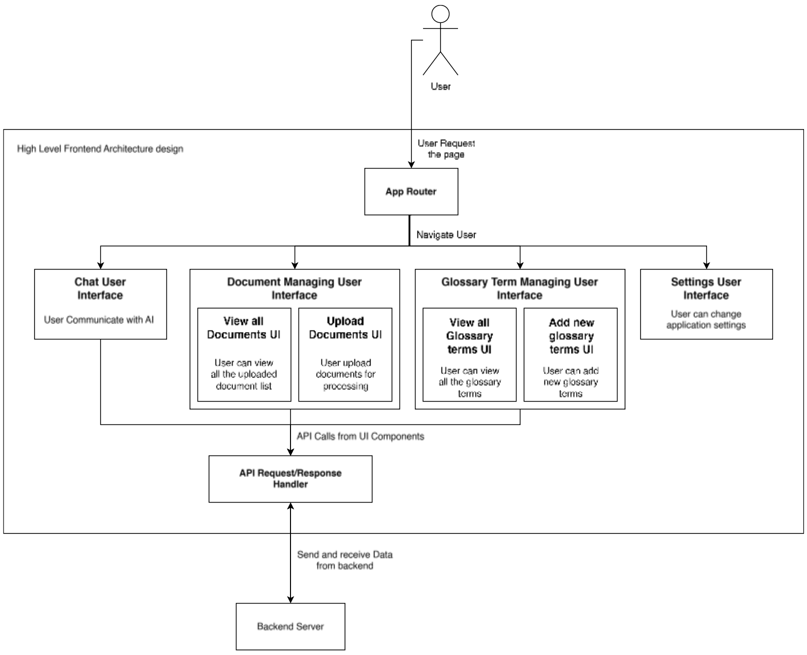

Figure 3: Frontend architecture

5.3 Backend Architecture

The backend architecture of the system is designed as a modular, service-oriented structure, with the core service being the Knowledge Graph Backend. This service handles critical operations such as document ingestion, storage, metadata management, knowledge graph updates, query processing, and response generation. When a user uploads a document via the frontend, the backend service securely uploads the file to Azure Blob Storage, while simultaneously storing document metadata. The uploaded document is then processed by the LLM, which identifies entities and relationships and updates the Neo4j-based knowledge graph accordingly. When a user submits a question, the query string is passed to this backend service, which extracts relevant entities and relationships, and performs semantic search in the knowledge graph to retrieve contextually related information. Additionally, the system fetches glossary definitions from the glossary database using fuzzy matching and retrieves past chat history associated with the document to enhance contextual understanding. Then user question, graph context, glossary definitions, and chat history compiled into a prompt and sent to the LLM. The LLM then generates a domain-specific response, which is returned to the frontend for user display.

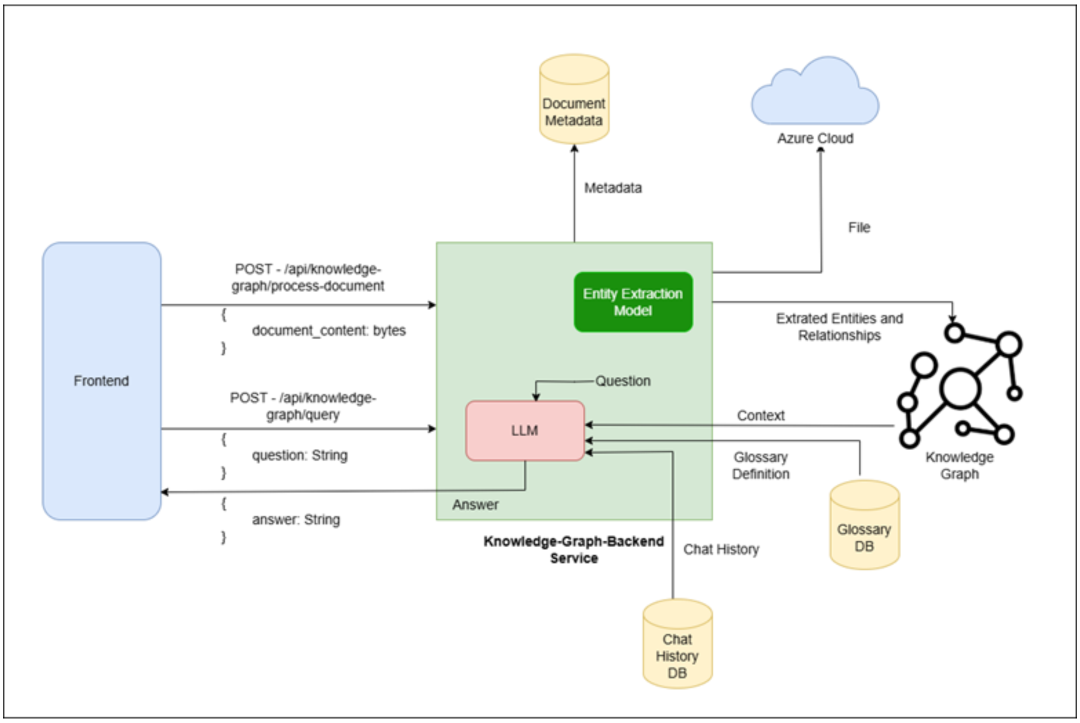

Figure 4: Backend architecture

5.4 Object-Oriented Analysis and Design (OOAD) Diagrams

As part of the system's design methodology, Object-Oriented Analysis and Design (OOAD) principles were applied to model the software components, their responsibilities, and interactions.

5.4.1 Class Diagram

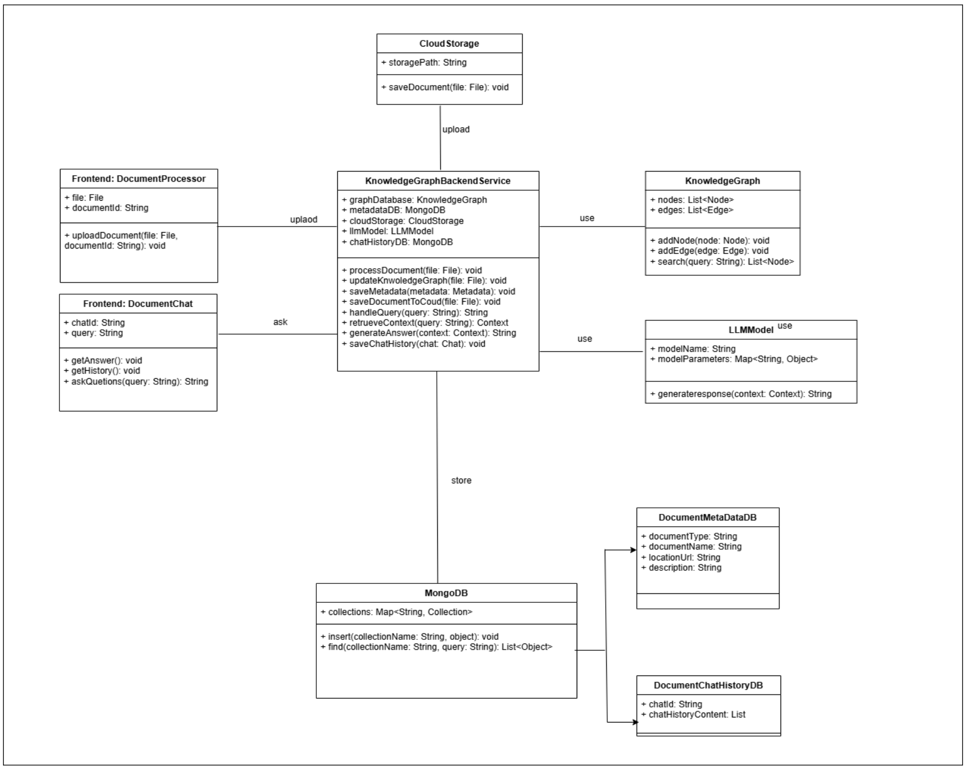

Figure 5: Class diagram

5.4.2 Sequence Diagram

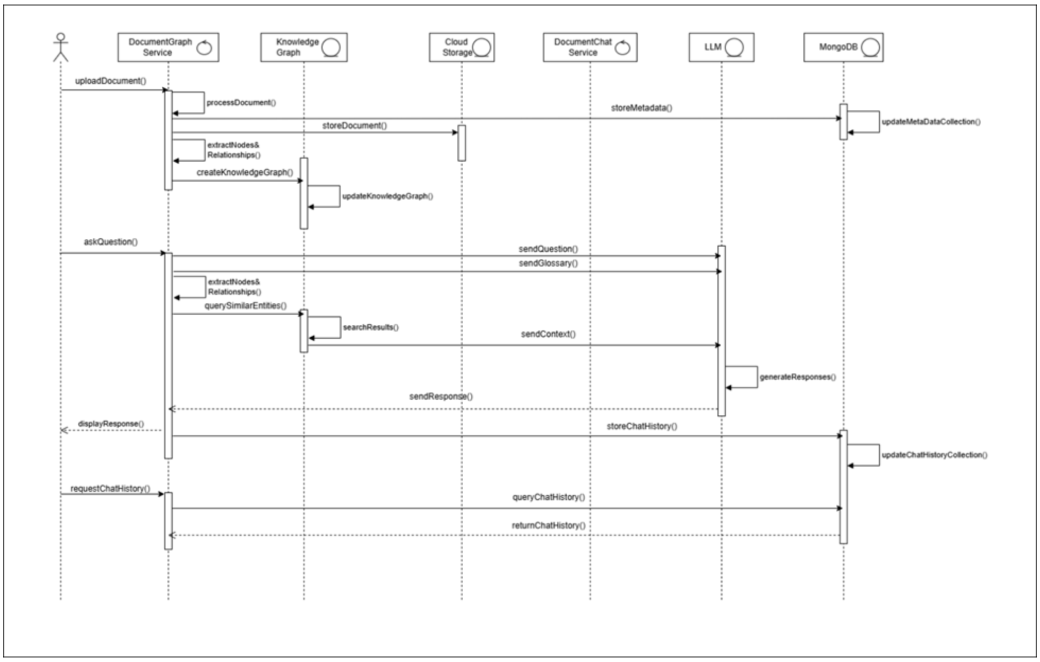

Figure 6: Sequence diagram

5.5 Database Design

The system incorporates a hybrid database design that utilizes both a NoSQL document-oriented database (MongoDB) and a graph database (Neo4j). This combination supports flexible metadata storage and efficient relationship traversal, respectively. Each database is purposefully selected to handle distinct aspects of the system, contributing to scalability, performance, and data consistency.

5.5.1 MongoDB Data Models

Metadata Collection

- The Metadata Collection in MongoDB is responsible for storing detailed information about each uploaded document. Each entry includes fields such as a unique id, filename, numberOfPages, hash, fileSize, fileType, uploadedDate, azureBlobUrl and azureBlobName. The hash field ensures document uniqueness for each user by identifying a preventing duplicate upload.

Figure 7: MongoDB metadata collection

Glossary Collection

- The Glossary Collection is designed to hold domain-specific terms and their corresponding definitions as maintained by each user. This collection enables the system to support semantic enrichment during question answering by retrieving definitions for technical terms that might not be understood by the language model. Each glossary entry includes an id, the term, its definition, and a createdAt timestamp.

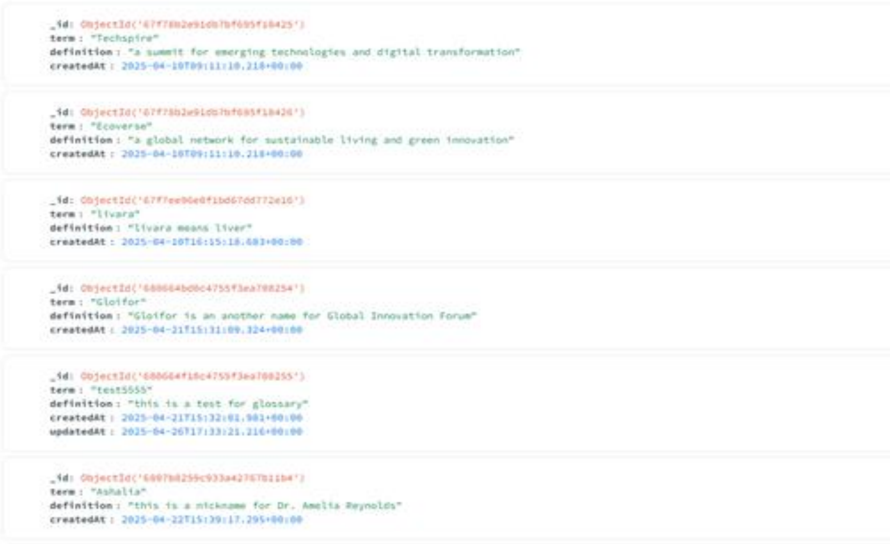

Figure 8: MongoDB glossary collection

5.5.2 Neo4j Knowledge Graph Model

The Neo4j Knowledge Graph stores entities and relationships extracted from user documents in a graph structure, enabling efficient semantic search and contextual understanding. Each document's key terms are represented as nodes, with relationships linking them based on their contextual meaning. When users upload a document, the graph is automatically updated. During question answering, the system uses the graph to retrieve relevant context by matching entities from the query. This context is then combined with glossary definitions and passed to the LLM to generate accurate responses.

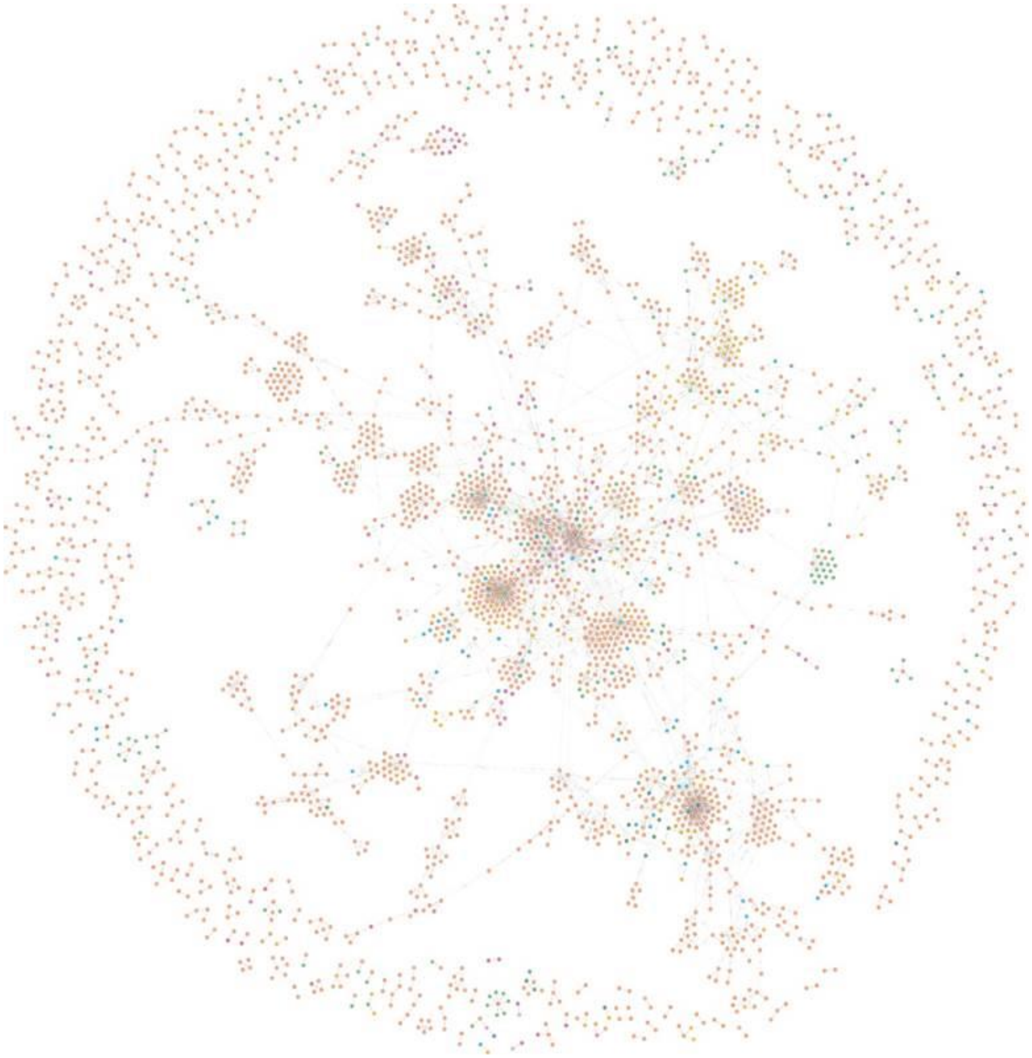

Figure 9: Visual representation of the knowledge graph

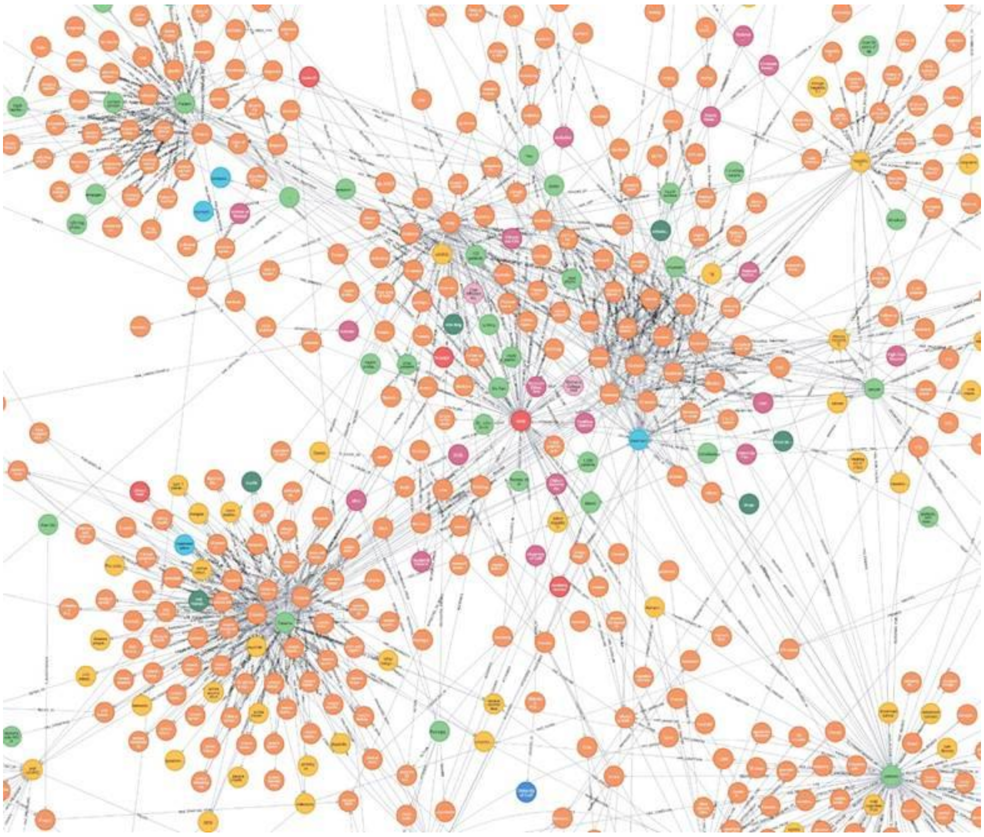

Figure 10: Zoomed-in view of the knowledge graph

6. Results and Evaluation

This chapter presents a comprehensive analysis of the system's performance in alignment with the research objectives and hypotheses. Evaluation was carried out in two key areas—retrieval performance and generation quality, followed by a focused assessment on the impact of domain-specific glossary integration. Each section includes a description of the test methodology, key findings, and interpretation in relation to the research hypotheses (H1 and H2).

6.1 Retrieval and Generation Evaluation (H1)

6.1.1 Retrieval Evaluation Metric and Test Procedure

Unlike traditional document retrieval—where retrieved text chunks or documents are evaluated—this work retrieves knowledge graph triples of the form *(head-node, relationship, tail-node)*. We measured context recall as our primary metric:

$$\text{Recall} = \frac{\text{Number of Relevant Triples Retrieved}}{\text{Total Number of Relevant Triples in Ground Truth}}$$

Precision and F-scores were de-emphasized because missing a relevant triple (false negative) poses a greater risk to downstream reasoning than including extra, potentially irrelevant triples (false positives). Given exhaustive matching based on entity overlap, precision is naturally low; recall more accurately reflects our system's goal of exhaustive context gathering.

6.1.2 Generation Evaluation Metrics and Test Procedure

This phase of evaluation focused on the quality of responses generated by the LLM when supported by retrieved graph triples. We evaluated two core aspects:

- Answer Relevance: Does the generated answer directly and meaningfully respond to the user's query?
- Answer Faithfulness: Does the answer stay true to the information in the retrieved triples?

Each metric was rated manually using the following scales:

Answer Relevance (Scale: 0 to 2)

- 0 = Not relevant at all
- 1 = Somewhat relevant (partial)
- 2 = Highly relevant (fully useful)

Answer Faithfulness (Scale: 1 to 5)

- 1 = Completely incorrect (hallucinated)
- 2 = Mostly incorrect
- 3 = Partially correct (some errors or fabrications)
- 4 = Mostly correct, but with minor inaccuracies
- 5 = Fully faithful and factually correct

Scores were then normalized to a 0–1 scale to ensure consistency.

6.1.3 Dataset and Manual Evaluation

A bespoke dataset of question–answer pairs was constructed. For each pair, we recorded:

- The ground truth triples manually annotated from source documents.
- The retrieved triples returned by our Graph Retriever.
- The generated response (used later in generation evaluation).

Because automated string-match evaluation cannot capture semantic equivalence (e.g., synonyms or paraphrased relations), all triple comparisons were performed manually—a common challenge in knowledge-graph evaluation.

6.1.4 Retrieval and Generation Evaluation Results

Table 1: Retrieval and Generation evaluation results

Metric	Mean	Median	Min	Max
Context Recall	0.839	1.000	0.000	1.000
Answer Relevancy	0.767	1.000	0.000	1.000
Faithfulness	0.933	1.000	0.600	1.000

- The mean context recall of 0.839 indicates that, on average, the system retrieved approximately 84% of all relevant triples required to answer a question.
- A median recall of 1.000 means that in at least 50% of the cases, the system successfully retrieved all ground truth triples, reflecting excellent consistency.

- The system achieved a mean Answer Relevance of 0.767, with a median of 1.000. This shows that in over 76% of the cases, the model generated responses that were fully or mostly relevant, and half of the queries achieved perfect relevance.
- Answer Faithfulness had a mean of 0.933 and a median of 1.000. Most responses stayed close to the retrieved facts, with only a small number introducing minor factual drift, usually due to overgeneralization or model assumption when the input context lacked complete detail.

6.1.5 Critical Interpretation of Results with H1

H1: "The implementation of a knowledge graph-based RAG system significantly improves the efficiency of information retrieval compared to traditional manual methods of document management."

The evaluation results strongly support this hypothesis. The improvements were observed not only in the retrieval phase but also in the quality of generated responses, which directly benefited from the structured and semantically enriched context provided by the knowledge graph.

Retrieval evaluation results interpretation

Compared to manual document browsing or conventional keyword-based search methods, the knowledge graph–based retrieval system demonstrated substantial advantages:

- Higher semantic precision was achieved by leveraging structured entities and relationships. This allowed the system to fetch only contextually relevant triples rather than vague or overly broad text blocks.

- Improved efficiency was observed across most test queries, with the majority returning accurate results in under a second. This contrasts with manual retrieval, which is often time-consuming and prone to human oversight.
- Better context granularity was ensured through the use of atomic triples. This reduced irrelevant information passed to the LLM, enabling more targeted reasoning during the generation phase.

Minor retrieval issues, where present, stemmed mainly from

- Incomplete or incorrect triple extraction during the initial graph construction process.
- Semantic variability, where equivalent meanings were expressed in different textual forms not captured during triple extraction.

Despite these, the retrieval module consistently delivered high recall and relevant facts, validating the efficiency claim of H1.

Generation evaluation results interpretation

The impact of improved retrieval was directly reflected in the generation quality. When accurate and complete triples were retrieved:

- The LLM consistently produced faithful answers that stayed true to the provided facts. This minimized the risk of hallucination—a common issue in LLMs when context is vague or ambiguous.
- The answer relevancy also improved, as the LLM had a focused and semantically rich context to interpret the question correctly and formulate responses accordingly.

Together, the strong performance in both retrieval and generation phases confirms that the system significantly enhances the overall information retrieval pipeline.

- It reduces user cognitive load by automating the search, filtering, and synthesis of relevant data.
- It increases the accuracy, speed, and interpretability of responses when compared to manual approaches or less structured retrieval systems.

Thus, both retrieval accuracy and generation fidelity complement each other in reinforcing the claim made by H1, clearly demonstrating the effectiveness of the knowledge graph–based RAG system for efficient and reliable information retrieval.

6.2 Glossary Integration Evaluation (H2)

To evaluate this Glossary integration, the system's performance was assessed in scenarios where questions contained domain-specific terminology. The glossary integration mechanism was designed to enhance the LLM's comprehension by supplying user-managed definitions for unfamiliar or domain-specific terms, thereby improving response relevance.

6.2.1 Evaluation Metric and Test Procedure

The evaluation of glossary integration focused specifically on the answer relevancy metric, as the primary goal was to assess how well the LLM could generate contextually appropriate responses when aided by glossary-provided definitions.

A controlled testing procedure was followed as below.

- A subset of the evaluation dataset was modified such that one-third of the questions included domain-specific made-up terms. These terms were designed to be semantically meaningful within the context but were intentionally invented to ensure they lacked any prior grounding in the LLM’s pretrained knowledge.
- Two versions of the system were tested
    1. Without Glossary Integration: In this setup, the system attempted to answer the glossary-based questions using only the context retrieved from the knowledge graph, without any access to glossary definitions.
    2. With Glossary Integration: Here, the glossary module was activated, allowing the system to incorporate domain-specific definitions into the context before response generation.
- For each question, the generated answers were manually scored for answer relevancy on a scale from 0 to 2, using the same criteria outlined in previous evaluation.
- The score was normalized by dividing the score by the maximum possible score (2), yielding a relevancy score between 0.0 and 1.0.

Manual evaluation was necessary due to the complexity of assessing semantic appropriateness and contextual accuracy, particularly when dealing with fabricated or domain-specific terminology.

6.2.2 Evaluation Results

Table 2: Glossary integration evaluation results

Answer Relevancy	Mean	Median	Min	Max

Without Glossary	0.483	0.500	0.000	1.000
With Glossary	0.717	1.000	0.000	1.000

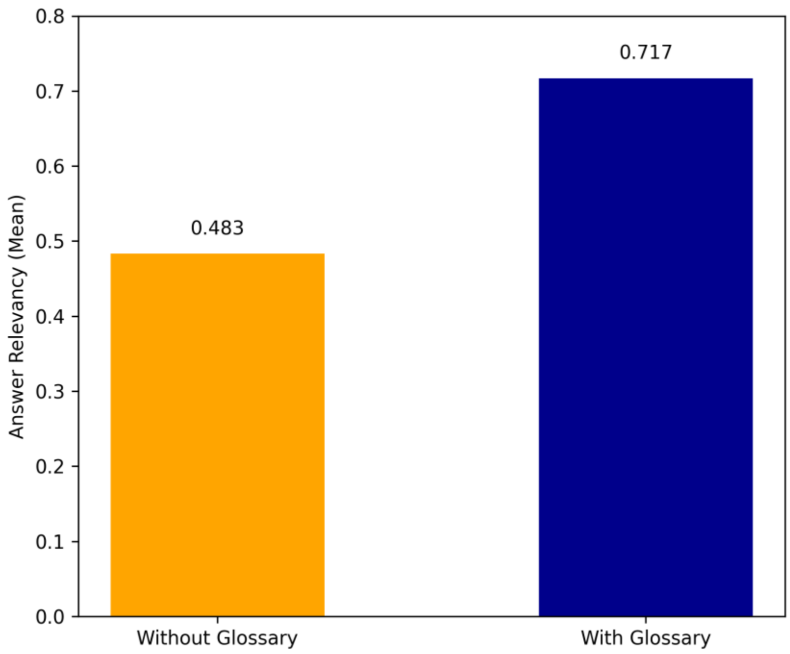

Figure 11: Bar chart of glossary integration comparison

- There was a 48.4% increase in the mean answer relevancy score after integrating the glossary (from 0.483 to 0.717). This clearly indicates that the glossary significantly contributed to improving the model's contextual understanding of domain-specific terms.

- The median score jumped from 0.500 to 1.000, showing that in over 50% of the glossary-based queries, the model was able to produce fully relevant answers when glossary support was provided.

6.2.3 Critical Interpretation of Results with H2

H2: “The integration of a domain-specific glossary enhances the contextual relevance and interpretative accuracy of responses generated by the LLM. This leads to more relevant answers by significantly improving the LLMs's contextual understanding.”

The observed results offer strong support for H2. In scenarios where the system faced domain-specific vocabulary, its baseline performance (without glossary support) showed considerable limitations. The average answer relevancy score dropped to 0.483, and the model frequently returned generic fallback responses such as:

“I'm sorry, I didn't understand your question. Could you please rephrase it?”

This behavior was expected, as the system prompt explicitly instructed the LLM to abstain from answering if insufficient context was available—an important measure implemented to avoid hallucination. This behavior validates that the lack of understanding stemmed not from flaws in the model itself, but from a genuine lack of contextual information.

Once the glossary module was introduced, the model was able to interpret the made-up terms correctly and generate more relevant answers. The answer relevancy improved significantly, reaching a mean score of 0.717 and a median of 1.0—a marked enhancement from the baseline.

This demonstrates the following key findings.

- Improved contextual grounding: The glossary definitions allowed the model to better understand the meaning of unknown domain-specific terms, enriching its internal context.
- Higher interpretative accuracy: Responses were more aligned with the intent of the question, particularly in cases where the glossary term played a central role in the query.
- Reduction in fallback responses: The frequency of “I don’t understand” type answers reduced dramatically after glossary integration, showing the glossary’s effectiveness in bridging knowledge gaps.

However, it is worth noting that the glossary-enhanced system still scored slightly lower than the original setup (without domain-specific terms in the questions), which had a relevance score of 0.767 in the previous evaluation. This marginal drop is likely due to the inherent limitations of the LLM’s in-context learning ability—while it can leverage glossary definitions effectively, it may not always fully incorporate them with the same fluency and nuance as natively learned language concepts.

Despite this, the results are highly promising. The experiment validates the usefulness of glossary integration for enhancing response quality in specialized domains. Furthermore, as LLMs continue to evolve and improve in their handling of in-context information, the effectiveness of this approach is expected to increase. The modular design of the glossary system also ensures easy scalability and adaptability across domains with minimal overhead.

7. Discussion

The findings and evaluation from this research project validate the effectiveness of the proposed Knowledge Graph-based Retrieval-Augmented Generation (RAG) system in enhancing domain-specific information extraction and query response accuracy. This discussion interprets the significance of the results with respect to the research objectives, hypothesis, and broader implications for knowledge management in institutions handling extensive, unstructured document archives. The primary goal of the system was to overcome the limitations of traditional document management methods, where decades of legal and regulatory documents exist in unstructured formats. The integration of knowledge graphs, large language models (LLMs), and glossary-driven contextual enhancement achieved this goal by enabling more precise, semantically aware retrieval and response generation.

The evaluation conducted under Hypothesis (H1) clearly demonstrated that the use of a knowledge graph for context retrieval significantly improved the system's ability to access and synthesize relevant information. The retrieval module achieved high mean context recall of 0.839, with perfect recall observed over 50% of cases. This confirms the robustness of the graph-based retrieval strategy in providing the LLM with comprehensive context for generating accurate responses. Additionally, answer relevancy and faithfulness metrics (0.767 and 0.933 respectively) underline the effectiveness of the prompt construction approach and reinforce that LLM generated responses remain grounded in retrieved context, minimizing hallucinations.

Furthermore, the integration of a domain-specific glossary– evaluated under Hypothesis 2 (H2) proved to be a critical addition. When the glossary module was enabled, the answer relevancy score rose from 0.483 to 0.717, representing a nearly 48% improvement. This indicates that glossary-aided retrieval meaningfully enhanced the model's understanding of domain-specific terminology, especially in instances where pretrained LLMs lacked familiarity with industry-specific language. The glossary's modularity and fuzzy matching mechanism ensured that it could dynamically support evolving domain lexicons with minimal user intervention. By allowing end users to manage and expand the glossary through the frontend, the platform remains adaptable to institutional changes and the introduction of new terminologies over time. This further contributes to the system's long-term sustainability and usability.

From a commercialization perspective, this project has strong potential to be developed into a viable product for industries that rely on legacy documents and require rapid access to institutional knowledge. Sectors such as law, finance, government, research, and healthcare, where large volumes of historical records are stored in PDF or scanned formats– could benefit significantly from this solution. The system architecture is designed with scalability and modularity in mind, leveraging widely adopted technologies such as Neo4j, MongoDB, Flask, React, and Azure Blob Storage. These choices support ease of deployment, cloud compatibility, and cost-effective operation.

To bring this solution to market, a commercialization process could involve productizing the web application as a SaaS platform, where clients can securely upload their document repositories, define their glossary, and use the conversational interface for intelligent information retrieval. Key considerations for market readiness include enhancing the user interface for broader accessibility, implementing multi-tenant data separation, offering integration with enterprise document management systems, and ensuring data privacy through role-based access control. Manufacturing or production costs would primarily involve cloud hosting, inference API subscriptions (e.g., Groq or DeepInfra), and support infrastructure, which can be optimized for commercial scale.

This research achieved its technical and academic objectives while also demonstrating the practical viability of the system as a commercial product. Its ability to bridge unstructured content with intelligent, structured access positions it as a valuable tool for any organization seeking to modernize its knowledge management processes.

8. Conclusion

This research project was carried out over the course of two semester and successfully achieved its main objective: the development of a knowledge graph-based retrieval-augmented generation system for domain-specific information extraction with glossary-aided responses that enables natural language access to institutional knowledge embedded in unstructured documents. The system combined knowledge graph-based context retrieval, a domain-specific glossary, and large language models to support accurate, context-aware responses to user queries. Through systematic planning and execution, the project covered problem identification, literature review, system design, implementation, and experimental evaluation, resulting in a fully functional web-based platform.

The system's performance, evaluated through two primary hypotheses, confirmed the effectiveness of using graph-based retrieval and glossary integration to improve the semantic understanding and relevance of LLM-generated answers. Key findings include a high context recall rate, strong faithfulness of responses, and notable improvement in answer relevance with glossary support. The modular architecture, built using widely adopted technologies such as Flask, React, MongoDB, and Neo4j, also demonstrated the system's scalability and readiness for real-world use.

Despite its success, the project faced several limitations. The LLMGraphTransformer, used for extracting triples from documents, sometimes failed to identify entity-relationship structures accurately, resulting in incomplete or incorrect knowledge graph construction. Additionally, processing time for triple extraction and graph updates was relatively high, making real-time document ingestion impractical. Another challenge was that even with the inclusion of glossary definitions, the LLM occasionally failed to interpret the intended meaning correctly, which highlights the dependency on the in-context learning capabilities of current LLMs.

In terms of future work, enhancing the precision of context retrieval is a key area for improvement. The current system retrieves all related triples, which may introduce irrelevant context into the prompt. A mechanism to filter and rank the most relevant triples based on the user query could significantly improve the focus and accuracy of LLM responses. Additionally, refining the triple extraction process, optimizing document processing time, and experimenting with fine-tuned or domain-specific models could further improve system performance.

Overall , this project not only fulfilled its technical and research goals but also laid the foundation for continued development in both academic and commercial directions, offering a scalable, AI-assisted solution for intelligent information retrieval from unstructured data sources.

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10. Appendices

Appendix 1: Frontend User Interfaces

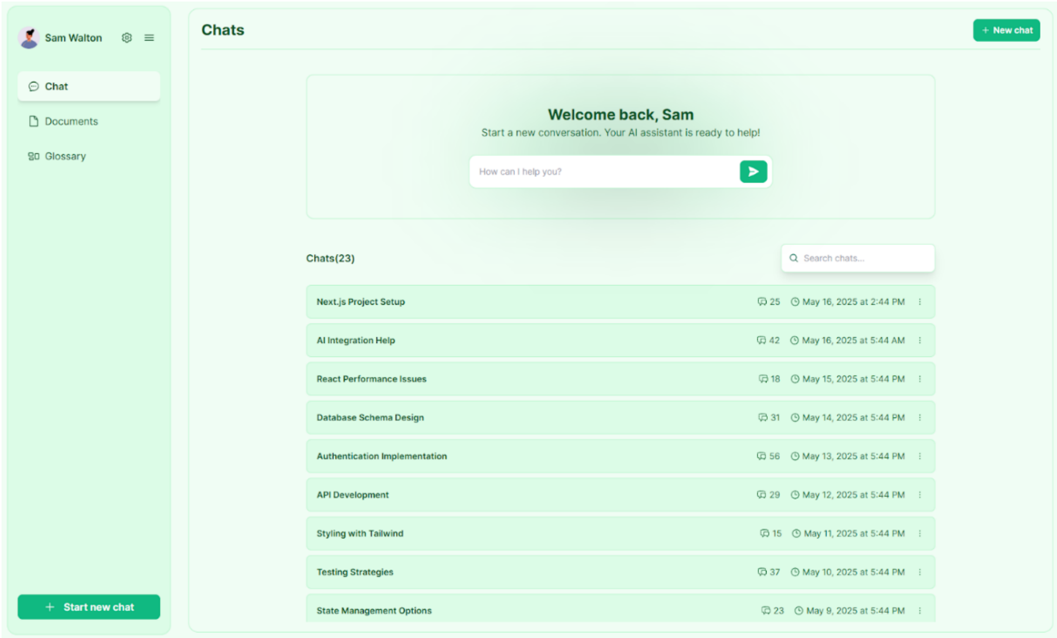

Figure 12: Welcome page of the webapp

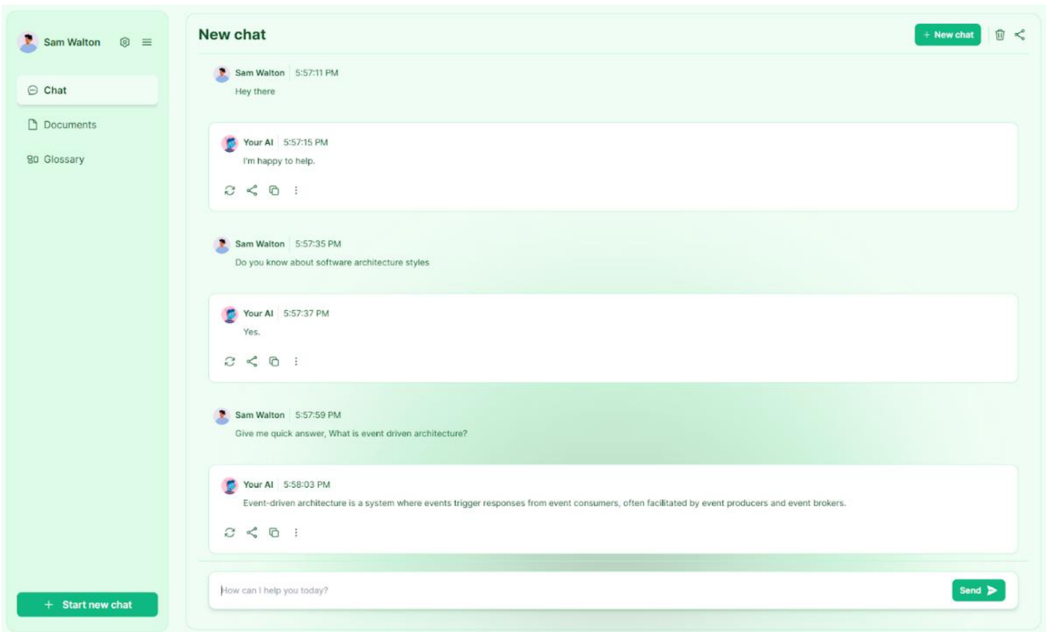

Figure 13: Chat interface

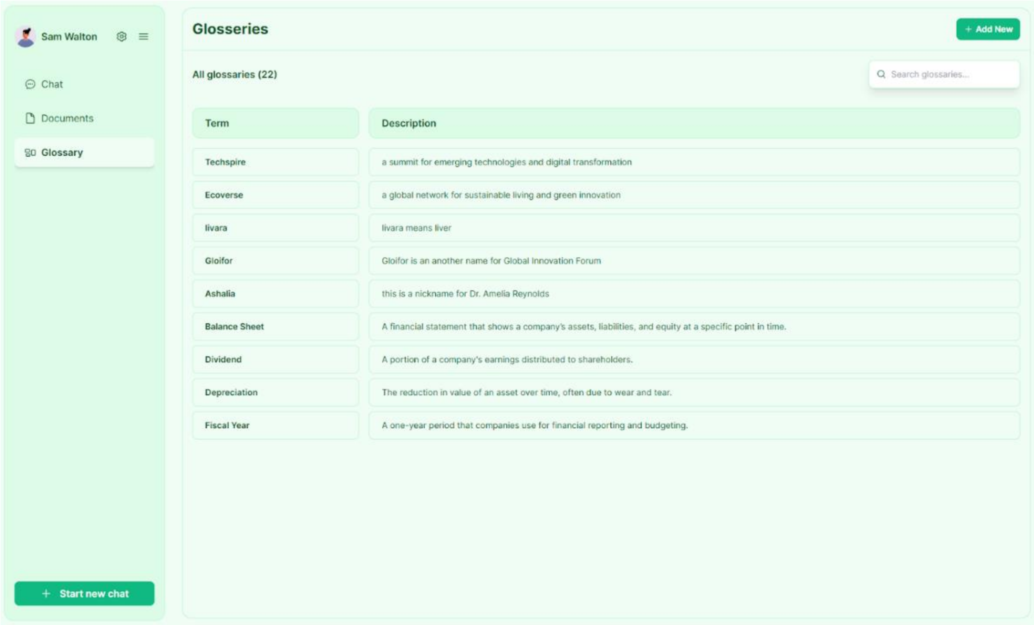

Figure 14: Glossary interface

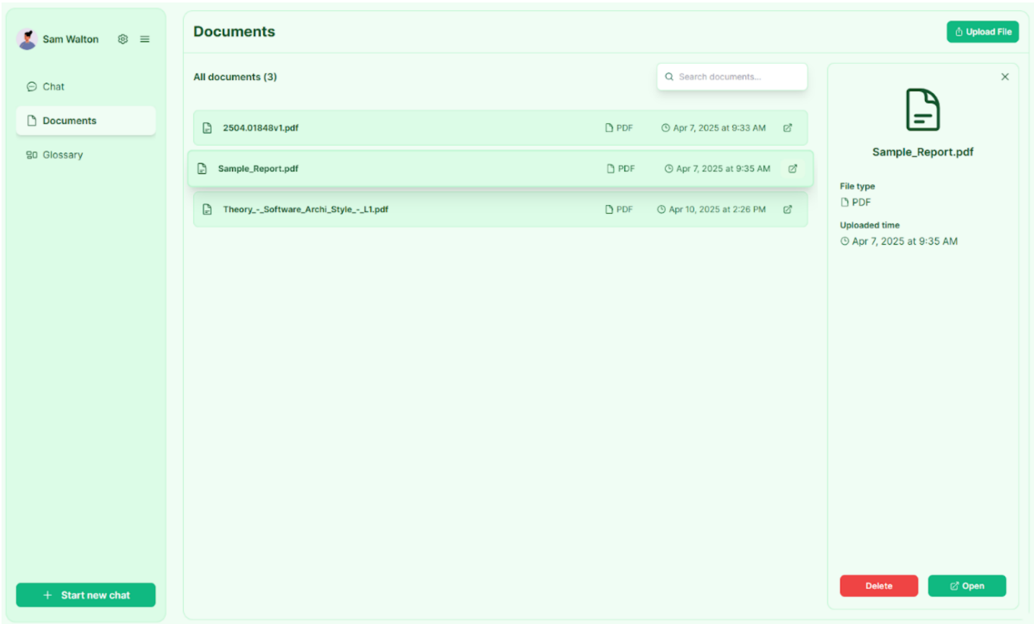

Figure 15: Document management portal

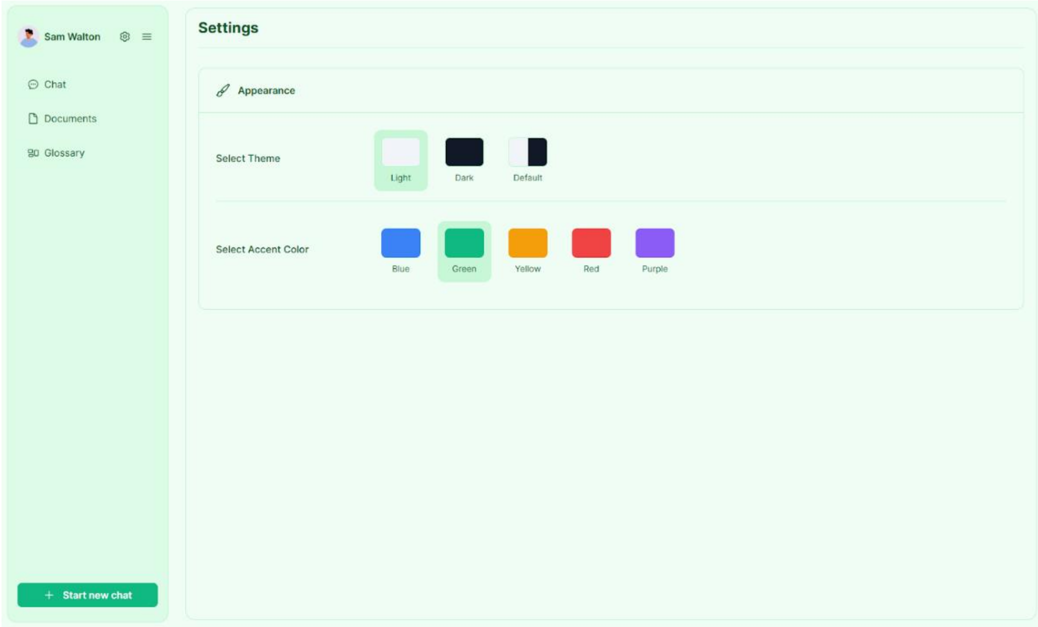

Figure 16: UI Theme 1

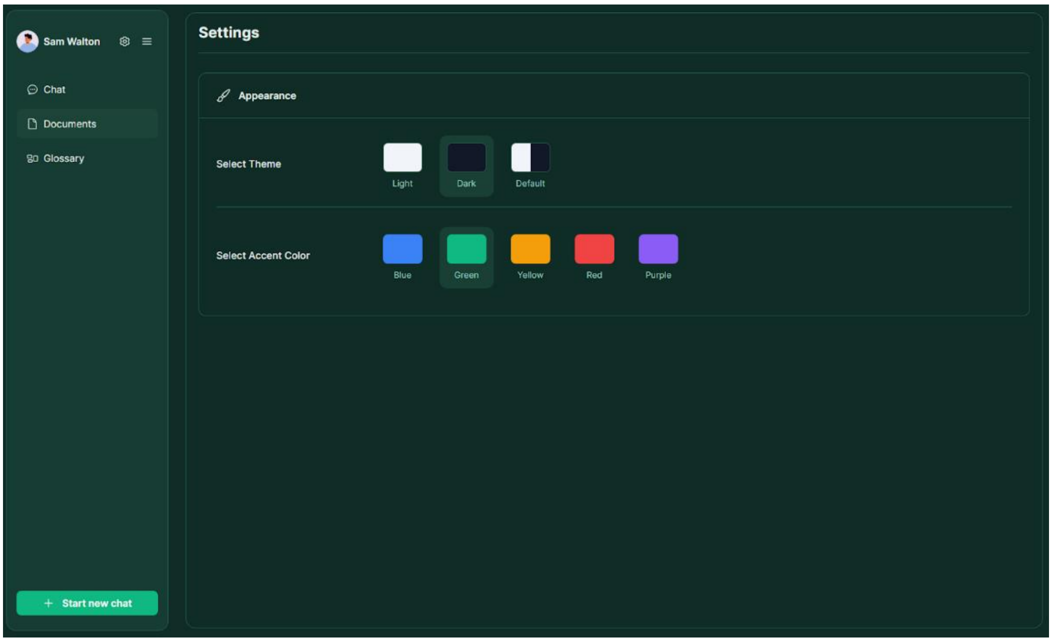

Figure 17: UI Theme 2

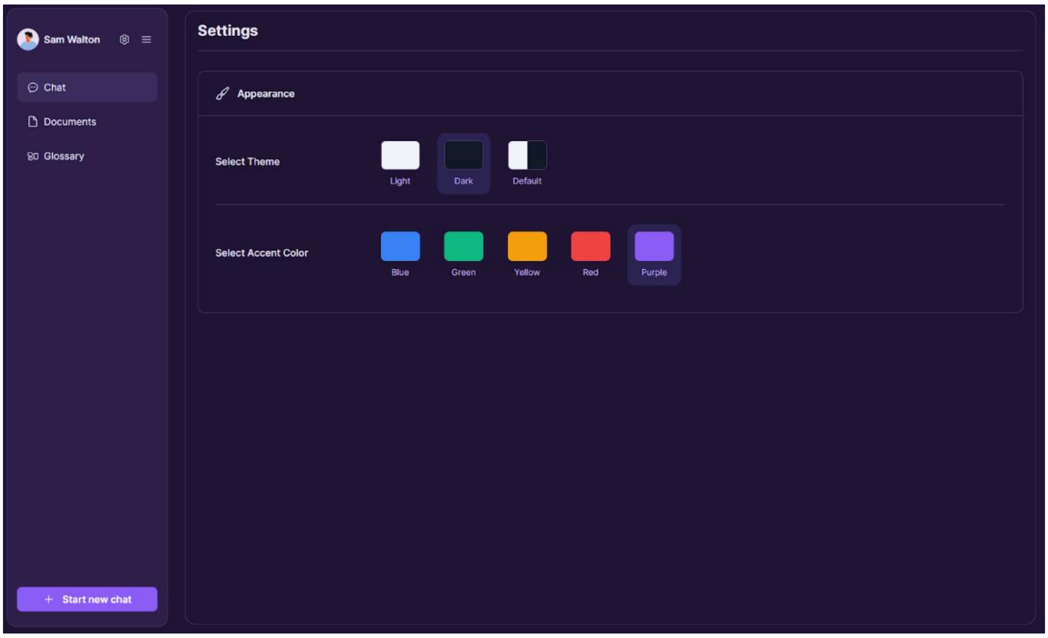

Figure 18: UI Theme 3

Appendix 2: Test Coverage and System Validation

Our system underwent comprehensive testing to ensure reliability and correctness. The test suite, implemented using pytest-cov, achieved an overall coverage of 85% across all the components. Critical modules including the knowledge graph handler (84% coverage), glossary handler (86% coverage), query handler(94% coverage) and answer generator (87% coverage) demonstrated robust validation of core functionality. The test suite comprised 534 statements across seven test files, covering essential operations such as graph construction, glossary term generation. The coverage report, generated using pytest-cov, revealed strong coverage in business logic components while identifying opportunities for improvement in infrastructure services.

Coverage report: 85%

Files Functions Classes

filter...

hide covered

coverage.py v7.8.0, created at 2025-05-14 12:46 +0530

File ▲	statements	missing	excluded	coverage
src\config\config.py	21	0	0	100%
src\config\logging_config.py	18	0	0	100%
src\generators\answer_generator.py	46	6	0	87%
src\handlers\glossary_handler.py	125	17	0	86%
src\handlers\knowledge_graph_handler.py	96	15	0	84%
src\handlers\query_handler.py	51	3	0	94%
src\main.py	103	31	0	70%
src\services\mongo_service.py	12	0	0	100%
src\services\storage_service.py	62	7	0	89%
Total	**534**	**79**	**0**	**85%**

coverage.py v7.8.0, created at 2025-05-14 12:46 +0530

Figure 19: Test coverage report

Appendix 3: Evaluation Spreadsheets

- **Retrieval and Generation Evaluation (H1)**

  https://docs.google.com/spreadsheets/d/1v67fuERPfk3DRztHyMrwn4T4WwzAr_qTKTs_ljVKnqU/edit?usp=sharing

- **Glossary Integration Evaluation (H2)**

  https://docs.google.com/spreadsheets/d/1_wGvBexw-E0EfIlCvr1qEgae2iFOc62a50xaNpQRGX8/edit?usp=sharing